

ME 300 Thermodynamics – Summer 2022

Exam 2

SOLUTIONS

(Part I) _____ (30 pts)

(Part II) _____ (40 pts)

(Part III) _____ (30 pts)

TOTAL _____

Directions

1. Open the online answers portion of the exam and download the exam (this document).
2. Make sure to show all your work!
 - a. *If you are using a tablet*, solve the problems directly on this document, showing all your work and writing as clearly as possible.
 - b. *If you are writing on paper*, solve the problems on a separate piece of paper (you do not have to print the exam), showing all your work and writing as clearly as possible.
3. Questions that require answers to be submitted to the online answers page on Canvas are written in *red italics* below as a way to remind you of what should be entered. All sketches and plots should be clearly drawn on your working paper and submitted with the upload.
4. If you have questions, please use the private chat function in Zoom to send a message directly to Dr. O'Connor so as not to disturb the other students. **DO NOT UNMUTE AND ASK YOUR QUESTION ALOUD** – this will be distracting to the other students.
5. Once the answers portion of the exam closes, scan or take very nice pictures of your work and upload it to the upload portion of the exam in a PDF document. I trust that you will not change anything on your exam after the answers portion of the exam closes. I can't enforce this – I'm really just relying on you to do the right thing. Please keep to the honor code.
6. The timeline for the exam will be the following (all times in EDT, Friday, July 22):
 - a. Exam opens (answers portion and upload portion): 11:00 AM
 - b. Answers portion of the exam closes: 12:30 PM
 - c. Upload portion of the exam closes: 12:45 PM
7. You can anonymously report any incidents of cheating using the following Google form:
<https://forms.gle/E4wFjDtvNuginVkP9>

Academic Integrity Pledge

I pledge that I am adhering to Penn State's Code of Conduct when taking this exam.

- I will not communicate with anyone during the exam except Dr. O'Connor.
- I will use only the allowed materials (calculator, notes, tables, and book), I will only access the internet to download the exam, view the e-book and notes on Canvas (no videos), and to submit my answers and upload the exam.
- I will not change any of my written answers after the exam but before I upload my work.
- I will not cheat in any way.

It is important to me to be a person of integrity and that means that all answers on this exam are my individual answers.

In the online answers, please type your full legal name to agree to the academic integrity pledge.

Part I: Concept Questions

1. (8 points) Can the thermal efficiency of a heat engine ever reach one? *Answer the multiple-choice question in Canvas.*

YES

NO +2

exact same question as practice exam

Explain your response to this question. *Type your answer in the box in Canvas.*

The second law, Kelvin-Planck statement says that a cyclically-operating device cannot completely convert heat to work. Since

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{in}} \text{ and } \dot{Q}_{in} > \dot{W}_{net} \text{ by 2nd law, } \eta_{th} < 1 \quad +3$$

2. (12 points) Identify the correct sign of the change in entropy for the following three processes.

- a) An ideal gas undergoes an isothermal increase in pressure. Is the change in entropy: *Answer the multiple-choice question in Canvas.*

GREATER THAN ZERO

LESS THAN ZERO +2

EQUAL TO ZERO

Same as HW8, #3 (a & b) and practice exam (c)

- b) An ideal gas undergoes isochoric temperature increase. Is the change in entropy: *Answer the multiple-choice question in Canvas.*

GREATER THAN ZERO +2

LESS THAN ZERO

EQUAL TO ZERO

- c) An ideal gas undergoes an adiabatic and irreversible expansion. Is the change in entropy: *Answer the multiple-choice question in Canvas.*

GREATER THAN ZERO +2

LESS THAN ZERO

EQUAL TO ZERO

- d) Identify the equations you used to explain the answers to these previous three questions. *Type your answer in the box in Canvas.*

$$\rightarrow a) \Delta S = C_p \ln(T_2/T_1) - R \ln(P_2/P_1) \rightarrow \text{since } T_1 = T_2 \text{ and } P_2 > P_1, \Delta S < 0 \quad +2$$

$$b) \Delta S = C_v \ln(T_2/T_1) + R \ln(V_2/V_1) \rightarrow \text{since } T_2 > T_1 \text{ and } V_1 = V_2, \Delta S > 0 \quad +2$$

$$c) \Delta S' = \oint \frac{\delta Q}{T} + \Phi \rightarrow \text{since } Q = 0 \text{ but } \Phi > 0, \Delta S' > 0 \quad +2$$

3. (10 points) Choose the device you would use to achieve each of the design goals below and describe why. *Answer the multiple-choice question in Canvas and then type your answer into the box in Canvas.*

a) Decelerate a flow of air before it enters a boiler system.

Same as HW7, #1

NOZZLE HEAT EXCHANGER

DIFFUSER COMPRESSOR

THROTTLE TURBINE

→ diffuser slows down flow due to expanding area

b) Control the flow of water into a tank.

NOZZLE HEAT EXCHANGER

DIFFUSER COMPRESSOR

THROTTLE TURBINE

Throttles control flow by dropping pressure

c) Increase both the temperature and pressure of air before it enters a processing unit.

NOZZLE HEAT EXCHANGER

DIFFUSER COMPRESSOR

THROTTLE TURBINE

Compressor raises pressure and temp by doing work on the fluid.

Part II: Short Problems

1. (20 points) Air is expanded in an isentropic turbine from an initial temperature of 1500 K and a pressure of 2 MPa to a final pressure of 0.1 MPa at a steady flow rate of 20 kg/s. Use the following properties for air to solve the questions below – $\gamma=1.4$ and $c_p=1001$ J/kg-K

Based on videos M6.9, M7.12, HW8, #4

- a) (8 points) What is the final temperature of the air at the exit of the turbine in [K]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- b) (8 points) What is the power produced by this turbine in [kW]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- c) (4 points) Draw this process on both a P-v and T-s diagram, labeling both states. *Draw your diagram below – do not enter anything into the Canvas quiz.*

Given : $T_1 = 1500$ K

$\gamma = 1.4$

$P_1 = 2$ MPa

$c_p = 1001$ J/kg-K

$P_2 = 0.1$ MPa

$\dot{m} = 20$ kg/s

Assume : air = ideal gas ^{+1/2}

$c_p = \text{const}$ ^{+1/2}

quasi-st. ^{+1/2}

Steady-flow ^{+1/2}

$\Delta S = 0$ ^{+1/2}

$\Delta ke = \Delta pe = \dot{Q} = 0$ ^{+1/2}

- a) Because ideal gas, $c_p = \text{const}$ and isentropic, can use isentropic relation:

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} = 1500 \left(\frac{0.1}{2} \right)^{\frac{0.4}{1.4}} \rightarrow T_2 = 637.34 \text{ K}$$

- b) Because steady flow: $\dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$

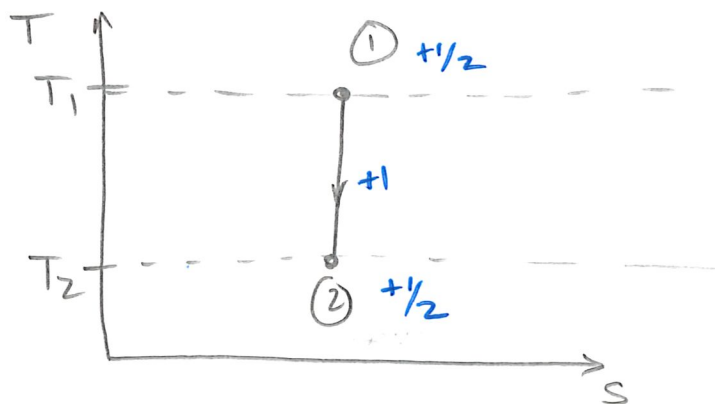
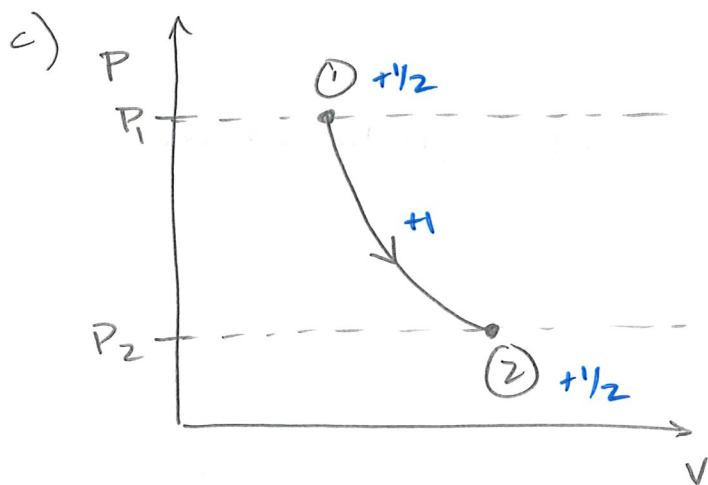
Because turbine: $\dot{m} \Delta h = -\dot{W}$

Because ideal gas w/ const. c_p : $\dot{m} c_p (T_2 - T_1) = -\dot{W}$

$$\dot{W} = -\dot{m} c_p (T_2 - T_1)$$

$$= -20(1001)(637.34 - 1500)$$

$$\dot{W} = 17.27 \text{ MW}$$



2. (20 points) A reversible Carnot cycle operates between 1500 K and 300 K with the following four steps:

copy of practice exam

- Process 1-2: isothermal heat addition
- Process 2-3: isentropic expansion
- Process 3-4: isothermal heat rejection
- Process 4-1: isentropic compression

- a) (3 points) What is the thermal efficiency of this cycle? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- b) (6 points) If 100 kJ of heat is added to the cycle in Process 1-2, how much heat was rejected in Process 3-4 in [kJ]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- c) (6 points) What is the change in entropy of the working fluid for each process in the cycle in [J/K]? *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- d) (5 points) Draw this cycle on a T-s diagram, labeling all the state points. *Draw your diagram below – do not enter anything into the Canvas quiz.*

Given: $T_H = 1500 \text{ K}$

$T_L = 300 \text{ K}$

Assume: ideal - reversible
: given eq.

a) $\eta_{th} = \eta_{max} = 1 - \frac{T_L}{T_H} = 1 - \frac{300}{1500} = \boxed{80\%}$ +1

b) $Q_H = 100 \text{ kJ}$

$\rightarrow Q_H = \bar{W} + Q_L$ +2

$Q_L = Q_H - \bar{W}$ where $\bar{W} = \eta_{th} Q_H = (0.8)(100) = 80 \text{ kJ}$ +2

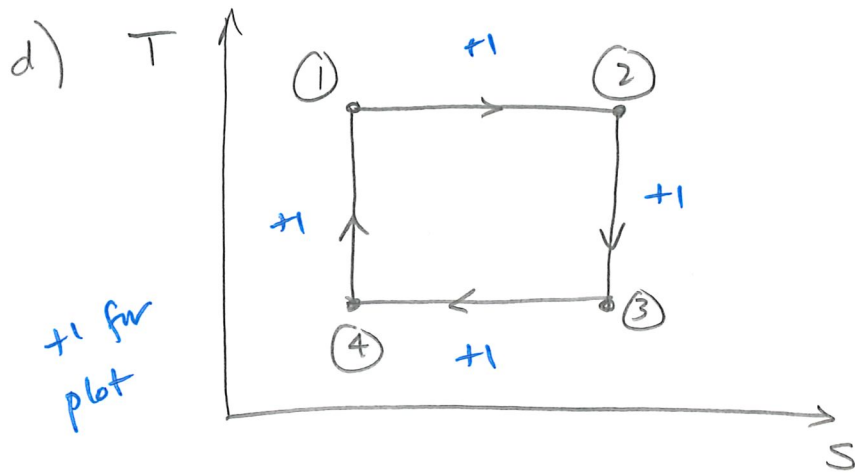
$Q_L = 100 - 80 \rightarrow \boxed{Q_L = 20 \text{ kJ}}$ +2

c) $S_2 - S_1 = \int \frac{\delta Q}{T} = \frac{Q_H}{T_H} = \frac{100}{1500} \rightarrow \boxed{S_2 - S_1 = 0.067 \frac{\text{kJ}}{\text{K}}}$ +1

$\boxed{S_3 - S_2 = 0 \text{ kJ}}$ +1 isentropic

$\boxed{S_1 - S_4 = 0 \text{ kJ}}$ +1 isentropic

$$\boxed{S_4' - S_3' = -0.067 \frac{\text{kJ}}{\text{K}}} \quad +1 \text{ by cycle definition}$$

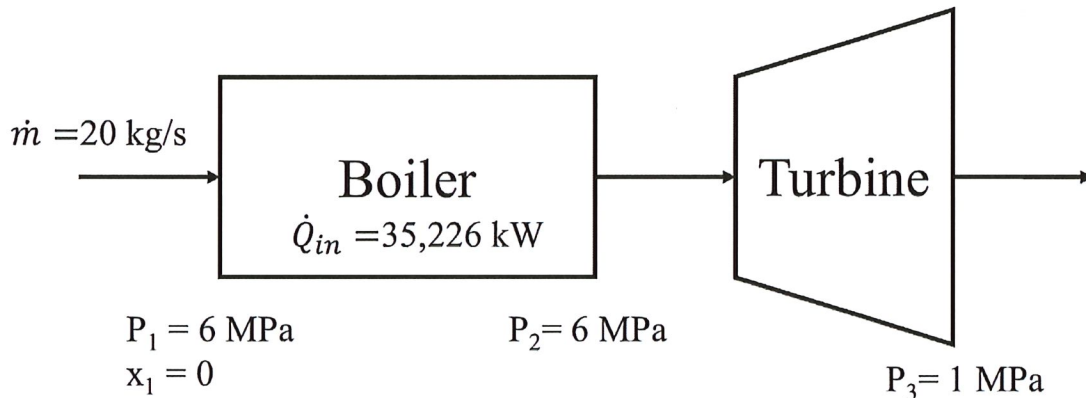


like HW6, #5

HN7, #3

Part III: Analysis (30 points)

The story: The West Campus Steam Plant at Penn State's University Park campus uses a set of boilers to produce steam to heat campus. Some of the high-pressure steam is kept at high pressure so that it can be piped all the way across campus, but some of it is expanded through a turbine to lower pressure so that it can be used at buildings next to the steam plant. In this problem, we're going to look at a situation where we don't need too much steam on campus (it's summer!) but we want to make more power. You're going to analyze the state of the water and calculate the power that can be produced by using a steam turbine at the steam plant. The diagram below shows the two components in this system – assume both components operate reversibly.



- (3 points) List the assumptions necessary to calculate the state points before and after the heat exchanger and the power generated by the turbine. *Type your answer in the box in Canvas.*
- (8 points) Calculate the temperature of the steam after the heat exchanger. Identify the table you used to calculate the state point. *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (4 points) Calculate the temperature of the steam after the turbine. Identify the table you used to calculate the state point. *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (10 points) Calculate the power produced by the turbine in [kW]. *Show your work below or on a separate page and enter this value in the Canvas quiz.*
- (5 points) Draw this process on a T-s diagram, labeling the state points and phases of fluid at each state. *Draw your diagram below – do not enter anything into the Canvas quiz.*

a) Assume: S.C.S. $+1/2$ Boiler: $\Delta ce = \Delta pe = \dot{W} = 0$ $+1/2$
 quasi-st. $+1/2$ Turbine: $\Delta ce = \Delta pe = \dot{Q} = 0$ $+1/2$
 steady-flow $+1/2$ reversible $+1/2$

b) Given: $\dot{m} = 20 \text{ kg/s}$
 $P_1 = 6 \text{ MPa}$ $P_2 = 6 \text{ MPa}$ $P_3 = 1 \text{ MPa}$
 $x_1 = 0$ $\dot{Q}_{in} = 35226 \text{ kW}$

* Steady-flow: $\dot{m}(\Delta h + \Delta \cancel{ke} + \Delta \cancel{pe}) = \dot{Q} - \dot{W} + 2$

Heat exchanger: $\dot{m}(h_2 - h_1) = \dot{Q}$

$$h_2 = h_1 + \frac{\dot{Q}}{\dot{m}} + 2$$

→ State 1: $P_1 = 6 \text{ MPa}$

$$x_1 = 0$$

$$h_1 = h_f = 1213.9 \frac{\text{kJ}}{\text{kg}} \xrightarrow{+1} \text{from Table D.2}$$

→ State 2: $P_2 = 6 \text{ MPa}$

$$h_2 = h_1 + \frac{\dot{Q}}{\dot{m}} = 1213.9 + \frac{35226}{20} = 2975.2 \frac{\text{kJ}}{\text{kg}} \xrightarrow{+1}$$

$$h_2 > h_g \rightarrow \text{vapor from Table D.2} \xrightarrow{+1}$$

$$\text{Table D.3N} \rightarrow \boxed{T_2 = 600 \text{ K}} \xrightarrow{+1}$$

c) State 3: $P_3 = 1 \text{ MPa}$

$$S_3 = S_2 = 6.2233 \frac{\text{kJ}}{\text{kg} \cdot \text{K}} \xrightarrow{+2} \text{Table D.3N}$$

$$S_f < S_3 < S_g \rightarrow \text{mixture} \rightarrow \text{Table D.2} \xrightarrow{+2, +1}$$

$$\boxed{T_3 = T_{\text{sat}} = 453.03 \text{ K}} \xrightarrow{+1}$$

d) Steady flow: $\dot{m}(\Delta h + \Delta \cancel{ke} + \Delta \cancel{pe}) = \dot{Q} - \dot{W} + 2$

Turbine: $\dot{m}(h_3 - h_2) = -\dot{W}$

$$\dot{W} = -\dot{m}(h_3 - h_2) + 2$$

$$\rightarrow \text{State 3: } x_3 = \frac{S_3 - S_f}{S_g - S_f} = \frac{6.2233 - 2.1381}{6.5850 - 2.1381} = 0.9187 \xrightarrow{+1}$$

$$h_3 = 2613.3 \frac{\text{kJ}}{\text{kg}} \xrightarrow{+1}$$

$$h_3 = h_g x_3 + (1 - x_3) h_f \xrightarrow{+1} = (0.9187)(2777.1) + (1 - 0.9187)(762.52)$$

$$\dot{W} = - \dot{m}(h_3 - h_2)$$

$$= -20(2613.3 - 2975.2)$$

$$\boxed{\dot{W} = 7238 \text{ kW}} \quad +2$$

